

SALSABILA RAFIFAH HANDIFA

Backend Developer | Fullstack Developer | QA Engineer | System Builder
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PROFESSIONAL SUMMARY

Computer Engineering graduate from Universitas Diponegoro with end-to-end experience across backend system development, quality assurance, and database-driven architecture. **Engineered production-grade systems from zero to deployment**, designing relational schemas, building 50+ REST API endpoints, writing automated test suites with 100% coverage across 53 files, and validating systems under 300+ concurrent users via Apache JMeter. **Detail-oriented across both code and data**: applies structured input validation, enforces data integrity at the database level, and executes systematic black-box and security testing before every production handoff. Won 1st Place at HackIn Fest 2025 (PT Sucofindo) as the sole technical developer and testing authority. A fast learner who quickly acquires new languages and frameworks, and thrives in structured, collaborative SDLC environments. Eager to bring ownership, accountability, and a testing-first mindset to your engineering team from day one.

EDUCATION

Bachelor of Computer Engineering	GPA: 3.82 / 4.00
Universitas Diponegoro — Semarang, Indonesia	2021 – 2025
Thesis: "Student Academic Progress Monitoring System at SDIT Darussalam 01 Batam: Backend Development"	
Relevant Coursework: Software Engineering, OOP, Database Systems, Cryptography, Computer Networks, Real-Time OS, IT Project Management, Algorithms & Programming	

SKILLS

Languages	JavaScript (ES6+), Go (Golang), Java, PHP (basic), SQL, HTML5, CSS3
Backend & API	Node.js, Express.js, Hapi.js, Gin, Fiber, RESTful API, Clean Architecture (4-layer), Microservices, API Design & Versioning
Frontend	React.js, Vue.js, Tailwind CSS, DOM Manipulation, Responsive UI
Database	PostgreSQL, MySQL, MongoDB, Redis, Schema Design, N+1 Query Optimization, Versioned Migrations, Connection Pooling, Data Integrity Enforcement
Testing & QA	Jest, Apache JMeter, Postman, Black-Box Testing, Bug Reporting, Regression Testing, Security Testing (JWT/RBAC scenarios), Unit, Integration, Functional, CI/CD-integrated Test Suites, Formal Test Documentation
Security & Auth	JWT Dual-Token (DB-backed revocation), RBAC Middleware, Argon2, Bcrypt, CORS, NGINX Rate Limiting, Signed URL Audit Trail
DevOps & Cloud	GitHub Actions (CI/CD), Docker, NGINX, Railway, AWS (Architect-level certified), RabbitMQ (async pipeline), Firebase Cloud Messaging (FCM)
Tools & Methods	Git, Azure DevOps, Agile/Scrum, SDLC, Postman, User Requirement Analysis, Technical Documentation, ERD Design
Soft Skills	Detail-Oriented, Bug Reporting, Analytical Thinking, User Requirement Analysis, Fast Learner, Cross-functional Communication, Delivery Under Pressure, Ownership & Accountability

EXPERIENCE

Fullstack Developer, Freelance PT Sinergi Informatika Semen Indonesia	Apr – May 2026
- Engineered a full FCM push notification backend using Lumen for PT Dahana's mobile e-Procurement platform, configuring per-action trigger logic across TMS and CMS modules and validating delivery across all authorized user roles prior to QA handoff. - Architected and integrated 2 Ad Hoc procurement modules (Commercial Evaluation and Awarding, each with list and detail views) with role-based permission configuration across Dev and QA environments, delivered. - Resolved 3 critical VMS bugs under active QA pressure: corrected a stale GET endpoint causing post-update data loss, added a missing Save action that blocked vendor profile submission, and implemented a 255-character live counter with visual warning on PR item description fields.	
Stack: React.js, PHP, Lumen, REST API, FCM/Firebase, Postman, CORS/Auth Validation, Agile/Scrum, Azure DevOps, Git	
Frontend Developer Intern PT Sinergi Informatika Semen Indonesia	Oct 2025 – Apr 2026
- Implemented a secure Blob session-scoped document delivery mechanism for PT LEN Industri's e-Procurement platform, replacing direct URL exposure across Evaluation Result and Technical Clarification endpoints and achieving 100% approval from Functional across all post-logout and unauthorized access test scenarios. - Sliced and integrated 8+ frontend modules across PT LEN Industri (VMS, Purchase Requisition, CMS) and PT Dahana (VMS, TMS) from Figma designs into API-connected React.js interfaces spanning the full procurement lifecycle from vendor registration through contract award. - Maintained 100% functional approval across 20+ deliverables on two concurrent BUMN e-Procurement platforms across 5+ Agile sprints, with zero rework requests and zero sprint rollbacks, through a systematic 1:1 API validation protocol covering response schemas, HTTP status codes, CORS headers, and auth token propagation against staging and production environments.	
Stack: React.js, Vue.js, REST API, Postman, CORS/Auth Validation, Agile/Scrum, Azure DevOps, Git	

- Built and delivered the **SAKTI platform as the sole fullstack developer** for a PT Sucofindo team competing in HackIn Fest 2025, a national open competition, resulting in **1st Place, Service Innovation Category**, with zero critical defects surfaced during the live competition demo.
- Performed full **user requirement analysis from stakeholder interviews**, produced ERD and API contract documentation, then independently designed a **10-table MySQL schema with 16 versioned migrations** and engineered **50+ REST API endpoints** with JWT/Argon2 dual-token auth, RBAC for 4 user roles, and 5 combinable middleware authorization strategies.
- Consolidated **142 inspection service records across 25+ BUMN units** into a single searchable, real-time platform with a **signed URL audit trail** (60-second per-request expiry) ensuring 100% access accountability on every sensitive document download, and integrated an **AI-powered chatbot** for self-service data access.

Stack: Node.js, Express.js, React.js, MySQL, Sequelize ORM, JWT, Argon2, RBAC, Cloudinary, Botpress AI, Postman

Backend Developer, Capstone Project | SDIT Darussalam 01 Batam

Aug 2024 – Jun 2025

- Designed a multi-role student academic monitoring system from zero. Conducted **user requirement analysis** across 4 stakeholder roles, translated requirements into a normalized **10-table relational MySQL schema**, and built 60+ REST API endpoints serving Admin, Teacher, Parent, and Student roles across web and Android clients, reducing academic reporting latency from a 4–6 month manual cycle to real-time access.
- Executed structured black-box testing across 100+ Postman test scenarios and 5 targeted JWT/RBAC security scenarios (no token, invalid token, expired token, valid token + correct role, valid token + mismatched role). Achieved **100% pass rate** and produced **formal test documentation** covering test cases, expected vs. actual results, and pass/fail conclusions for each scenario.
- Validated **zero critical failures under 300+ concurrent users** across load, stress, and spike testing (Apache JMeter). Deployed to Railway with cloud-hosted MySQL, connection pooling, and Redis rate limiting with MemoryStore fallback for graceful degradation under peak load.

Stack: Node.js, Express.js, MySQL, Sequelize ORM, JWT, Bcrypt, RBAC, Redis, Apache JMeter, Postman, Railway

Backend Developer Intern | PT Praisindo Teknologi — Jakarta Selatan

Jan – Apr 2024

- Delivered 2 production Go microservices (AML Watchlist Approval & Customer Risk Profile Detail) into a fintech compliance data pipeline, as measured by **zero blocker defects across 2 consecutive QA sprint handoffs**, by developing in Golang + MongoDB and executing structured Postman validation covering happy-path, null/invalid input, and authentication edge-case scenarios before every handoff.
- Authored API contracts, data flow documentation, and integration guides adopted as the primary technical reference by 3 cross-functional teams (Backend, Frontend, QA). Received **zero clarification requests** across 2 sprints in a 10+ person team tracked via Azure DevOps, demonstrating structured technical communication and **user requirement analysis** capability.
- Built two structured REST APIs in Go independently** from scratch. Lecturer API (Fiber, modular layered architecture) and ToDoList API (Gin, Clean Architecture 4-layer with interface-based dependency injection, GORM, connection pooling), mastering Go module structure before contributing to the production codebase.

Stack: Go (Gin, Fiber), MongoDB, PostgreSQL, GORM, Microservices, Clean Architecture, Postman, Azure DevOps, Git

Mobile Programming Practicum Assistant | Universitas Diponegoro

Aug – Dec 2024

- Mentored **168 undergraduate students** in advanced Android development using Kotlin (Module 2). Conducted hands-on code reviews and debugging sessions, reviewed 80+ assignments and practical exams, and supervised 16+ cross-platform final mobile application projects, achieving a **100% project submission** and completion rate aligned with academic and technical standards.

Stack: Kotlin, Android SDK, Mobile Development, Code Review, Technical Mentoring

KEY PROJECTS

Bookshelf App | Vanilla JavaScript · HTML5 · CSS3 · localStorage · DOM API

Apr 2026

- Built a fully browser based book collection manager from zero using Vanilla JavaScript with no framework, no library, and no backend dependency. Implemented unidirectional data flow (state to localStorage to DOM) as the core architectural pattern. Independently identified and resolved two classes of real world risk. First, XSS injection via user supplied input rendered into innerHTML, resolved by implementing an escapeHtml() sanitization function. Second, corrupted localStorage data causing application crashes on startup, resolved by wrapping JSON.parse in a try/catch block with a downstream data integrity filter that discards malformed records before they enter application state. Score 4/5, Dicoding.

Forum API | Node.js · Hapi.js · PostgreSQL · Jest · Docker · NGINX · GitHub Actions · JWT

Jan 2026

- Architected a production grade REST API across 4 Clean Architecture layers (Domain, Application, Interface, Infrastructure), fully decoupling all 16 use cases from infrastructure with each independently testable via mock repository injection. Identified and resolved an N+1 query bottleneck in comment like aggregation by replacing per row DB calls with a single batched IN(...) plus GROUP BY PostgreSQL query. Hardened all 15+ endpoints with JWT dual token auth, bcrypt hashing, DB backed token revocation, and NGINX rate limiting (90 req/min). Delivered 100% test coverage across 53 files with automated build-migrate-test-deploy via GitHub Actions CI/CD and containerized with Docker + NGINX for Railway deployment, eliminating all manual deployment steps. **Perfect Score 5/5**, Dicoding.

Bookshelf API | Node.js · Hapi.js · nanoid · ESLint · In-Memory Storage

Jan 2026

- Engineered a RESTful CRUD API from scratch with multi-parameter composable query filtering (title, reading status, finished status) and layered input validation (required field enforcement + logical consistency checks). Solved a data consistency problem by computing the finished field server-side as pageCount === readPage on every write, eliminating the possibility of conflicting state between stored fields. Enforced consistent code style across the codebase using ESLint with Dicoding's Airbnb-based configuration. **Perfect Score 5/5**, Dicoding.

- Reduced redundant PostgreSQL load on high-traffic playlist endpoints by integrating a Redis cache layer (30-min TTL), eliminating repeated DB hits per cache cycle and improving response latency. Decoupled email export from the API via a RabbitMQ async pipeline (producer/consumer, message acknowledgement), ensuring 100% non-blocking request handling for all export operations. Designed a normalized 10-table PostgreSQL schema with versioned migrations supporting RBAC and full activity audit logging across 8 resource domains. **Perfect Score 5/5**, Dicoding.

Geometry Calculator App | Java · Java Swing · Apache Ant · OOP

May 2022

- Designed and built a Java Swing desktop application from scratch for the DKP practicum final project, calculating area and perimeter for 8 geometric shapes via a graphical user interface. Solved the user input reliability problem by implementing JFormattedTextField with NumberFormatter, restricting all input fields to numeric only values and preventing runtime crashes from non numeric characters. Applied OOP principles throughout with each geometric shape as a dedicated JFrame class following Single Responsibility, consistent window lifecycle management via setVisible(true) plus dispose(), and multi exception handling (ClassNotFoundException, InstantiationException, IllegalAccessException, UnsupportedOperationException) for robust UI startup. Built and distributed as a standalone executable JAR via Apache Ant.

CERTIFICATIONS

- **Dicoding Indonesia:** Back-End Developer Expert with JavaScript (2026), AWS Cloud & Gen AI Fundamentals (2026), Architecting on AWS (2025), SOLID Principles in OOPS (2024), SQL Fundamentals (2023)
- **Pusat Pengembangan Talenta Digital:** Intermediate Junior Web Developer (2025)
- **Bangkit (Google, GoTo, Traveloka):** Mobile Development Learning Path (2024)
- **Oracle Academy:** Database Foundations & Design (2023)
- **Cisco Networking Academy:** Switching, Routing & Wireless Essentials (2023)

ACHIEVEMENTS

- **1st Place — HackIn Fest 2025**, Service Innovation Category (PT Sucofindo): Won nationally as the sole technical developer (backend + frontend) delivering a full-stack SAKTI platform with zero critical defects, selected as the best innovation from a national open competition, validating the ability to deliver a high-impact, flawlessly tested digital solution under intense deadline pressure.
- **Best Staff Award 2022 — AI Muharrik:** Recognized for exceptional initiative; subsequently promoted to Division Head, leading 7 structured programs with 100% on-time delivery.

ORGANIZATION EXPERIENCE

Head of Annisa Division | AI Muharrik — Semarang, Indonesia

Sep 2023 – May 2024

- Promoted to Division Head following Best Staff Award 2022; delivered **100% on-time completion** across 7 structured programs (5 internal, 2 cross-departmental) with 25+ participants per major event, managing full program lifecycle from planning and stakeholder alignment through execution and post-event reporting.

Member, Software Division | Computer Engineering Research Club — Semarang, Indonesia

Apr 2023 – Oct 2024

- Selected as 1 of 26 members from 46 applicants through a competitive screening process; contributed to a collaborative portfolio project delivered via a disciplined Git-based team workflow, mirroring professional software development practices.